Grade 8 Name____________________
Unit 3 Date____________________
Lesson 5 Practice Period____________________

Complete the table and then answer question 1.

Number	Standard Scientific Form
$4{,}500{,}000{,}000$	
45×10^6	
45×10^7	
45×10^5	

1. Which numbers in the table are represented by the same standard scientific form?

2. Find the sum by filling in the blanks.

$$(5.07 \times 10^6) + (4.5 \times 10^5)$$

$$(\_\_\_\_ \times 10^5) + (4.5 \times 10^5)$$

$$= (55.2 \times 10^{\boxed{}})$$

$$= (\_\_\_\_\_ \times 10^{\boxed{}})$$

Use the given information to answer questions 3-5.

The population of Gainesville was $141{,}085$ in 2020. The population of Jacksonville was $949{,}611$ in 2020.

3. How many more people lived in Jacksonville compared to Gainesville in 2020? Write your answer in scientific notation.

4. Write the population numbers from the word problem in scientific notation.

$141{,}085$	$\_\_\_\_\_ \times 10^{\boxed{}}$
$949{,}611$	$\_\_\_\_\_ \times 10^{\boxed{}}$

5. What is the difference between the populations? Write your answer using scientific notation.

6. Two cells are viewed under a microscope. Cell A has a diameter of 4.02×10^{-6} and Cell B has a diameter of 6.8×10^{-5}. Fill in the blanks to find the difference in diameters of Cell A and Cell B.

$$(4.02 \times 10^{-6}) - (6.8 \times 10^{-5})$$

$$(\_\_\_\_\_ \times 10^{-5}) - (6.8 \times 10^{-5})$$

$$= (\_\_\_\_\_ \times 10^{-5})$$

$$= (\_\_\_\_\_ \times 10^{\boxed{}})$$

7. Find the difference. Write your answer using scientific notation.

$$7.32 \times 10^{5} - 9.68 \times 10^{4}$$

8. Find the sum. Write your answer using scientific notation.

$$4.2 \times 10^{7} + 2.1 \times 10^{5}$$

The table lists the average mass, in kilograms (kg) of different types of large sea creatures. Use the information to answer questions 9 and 10.

Creature	Mass of One Individual (kg)
Blue Whale	1.9×10^{5}
Orca	9.9×10^{3}
Grey Whale	2.3×10^{3}

9. What is the difference in the mass of a blue whale and a grey whale?

10. What is the total mass of an orca and a blue whale?